

cointegration we consider two time-series v_t and m_t with common stochastic trends

$$v_t = n_{t-1} + \varepsilon_t^n + \varepsilon_t^v \quad (13.3)$$

$$m_t = n_{t-1} + \varepsilon_t^n + \varepsilon_t^m \quad (13.4)$$

where n_t is a RW process $\varepsilon_t^n, \varepsilon_t^v, \varepsilon_t^m \sim I(0)$ independent of each other and $v_t, m_t \sim I(1)$. Thus

$$v_t - m_t = \varepsilon_t^v - \varepsilon_t^m \quad (13.5)$$

therefore it can be said that v_t and m_t have a stochastic trend given by n_t and are therefore cointegrated. This result has important forecasting consequences: Given a pair of RW processes with a common stochastic trend, if a combination of them produces a stationary error/disequilibrium term then theoretically they can be predicted and exploited.

Dickey and Fuller (1979) demonstrated that if Equation 13.2 has a unit root, the process itself may be cointegrated where one or more combination of variables may be stationary though the individual variables may be non-stationary. Two variables which are cointegrated do not trend far away from one another. Thus cointegration is the long-term relationship between the series, where both are $I(1)$ but a linear combination of the two series is stationary. Given this framework we can describe a standard cointegration analysis for statistical trading:

1. Test for unit roots: usually done using the augmented Dickey-Fuller (ADF) test of order q which is based on the regression:

$$\Delta S_t + a + \beta S_{t-1} + \gamma_1 \Delta S_{t-1} + \dots + \gamma_q \Delta S_{t-q} + \varepsilon_t \quad (13.6)$$

where Δ is the first difference operator. The lags q are used to remove any autocorrelation which could introduce bias into the residuals. Use the t-ratio test statistic on $\hat{\beta}$ with $H_0: \beta$ vs $H_1: \beta < 0$.

2. If all series are confirmed to be stationary, test for cointegration. The simplest measure uses the Engle-Granger regression which is an ordinary least squares (OLS) regression:

$$S_t^1 = \beta_1 + \beta_2 S_t^2 + \dots + \beta_n S_t^n + \varepsilon_t \quad (13.7)$$

If the unit roots indicate that ε_t is stationary, then the variables $S^A \dots S^n$ are cointegrated with cointegrating vector $(1, -\hat{\beta}_2, \dots, -\hat{\beta}_n)$ or in other words $W = S_1 - \hat{\beta}_2 S_2 - \dots - \hat{\beta}_n S_n$. A portfolio is constructed off of this